Lab Session: Knowledge Base Construction, Alignment, and Expansion

(Preparation for Knowledge Graph Embedding)

Objective

In this lab, you will construct a private Knowledge Base (KB), align it with an open Knowledge Base (DBpedia or Wikidata), extend it via SPARQL queries, and prepare it for the next lab session on Knowledge Graph Embedding (KGE).

The final KB must be large enough to support meaningful embedding learning while remaining computationally manageable on a laptop.

An example toy data is provided for better explaination in each step.

Step 1 --- Build the Initial Private Knowledge Base

Requirements

You should have prepared your own knowledge graph in RDF model, with data collected from some web API or html pages. It is adviced that your starting knowledge graph respects the following volume:

- **≥ 100 triplets**
- **≥ 50 entities**
- No upper limit

Example (Domain: Scientific Researchers)

```
:MarieCurie :wonAward :NobelPrize .
:MarieCurie :fieldOfWork :Physics .
:AlbertEinstein :fieldOfWork :Physics .
:AlbertEinstein :wonAward :NobelPrize .
:RosalindFranklin :affiliatedWith :KingsCollegeLondon .
```

Good Practices

- Use URIs, not plain strings
- Normalize predicate naming (e.g., `camelCase`)
- Avoid duplicate triples
- Clearly separate entities from literals

Step 2 --- Entity Linking with Open Knowledge Bases

Requirement

For each entity:

- If it exists in DBpedia or Wikidata → link it.
- If it does not exist → create a new entity and define it semantically. (You can use Protégé for GUI. or purly with RDFLib)

You can use APIs mentioned in the CM. Mark also the confidence values return by the API. (As shown in the CM)

Deliverables

- Mapping table:

Private Entity	External URI	Confidence
:MarieCurie	wd:Q7186	0.99

- Ontological definition of newly created entities.
- Python script.

Example: Existing Entity

Private entity:

```
:MarieCurie
```

Wikidata match:

```
wd:Q7186
```

Alignment:

```
:MarieCurie owl:sameAs wd:Q7186 .
```

Example: New Entity (Not Found)

```
:QuantumResearchCenterParis rdf:type :ResearchInstitute .
:ResearchInstitute rdfs:subClassOf :Organization .
```

Define properties:

```
:affiliatedWith rdfs:domain :Person ;
                rdfs:range :Organization .
```

Step 3 --- Predicate Alignment Using SPARQL

Requirement

Use SPARQL queries on the open KB to determine whether:

- An identical predicate exists
- A semantically similar predicate exists

Validation can be manual. Here is a simple way: Given a triplet (s,p,o) in your private knowledge graph, the subject s_{1} and object o_{1} are alighed with {wiki(s)} and {wiki(o)} in the open KB, you can list all existing prdicates returned by the SPQRAL Query

```
SELECT ?property ?propertyLabel WHERE {
  ?property a wikibase:Property .
  ?property rdfs:label ?propertyLabel .
  {wiki(s)} ?property {wiki(o)}.
}
```

Then manually compare the results with p in your private knowledge graph, chose the most pertinent one for alignement.

Of course other more intelligent ways exist. You will learn later.

Below is another example of aligning by the text meaning

Example

Private predicate:

```
:wonAward
```

SPARQL Query (Wikidata)

```
SELECT ?property ?propertyLabel WHERE {
  ?property a wikibase:Property .
```

```
    ?property rdfs:label ?propertyLabel .
    FILTER(CONTAINS(LCASE(?propertyLabel), "award"))
    FILTER(LANG(?propertyLabel) = "en")
  }
  LIMIT 50
```

Candidate:

- `wdt:P166` → "award received"

If validated:

```
:wonAward owl:equivalentProperty wdt:P166 .
```

If broader/narrower:

```
:receivedRecognition rdfs:subPropertyOf wdt:P166 .
```

Step 4 --- KB Expansion via SPARQL

Final Size Requirements

After expansion, your KB must contain:

- **50,000 -- 200,000 triplets**
- **5,000 -- 30,000 entities**
- **50 -- 200 relations**

If the KB is too small → KGE models will fail to learn meaningful structure.
If too large → computation may exceed laptop capabilities.

Anchored Expansion Strategy

Only expand from confidently aligned entities.

1-Hop Expansion

```
SELECT ?p ?o WHERE {
  wd:Q7186 ?p ?o .
}
LIMIT 1000
```

Repeat for all aligned core entities.

Predicate-Controlled Expansion

```
SELECT ?s ?p ?o WHERE {
  ?s wdt:P166 ?o .
}
LIMIT 20000
```

2-Hop Example

```
SELECT ?award ?p ?o WHERE {
  ?person wdt:P166 ?award .
  ?award ?p ?o .
}
LIMIT 10000
```

Cleaning Before Embedding

Before exporting for KGE:

- Remove duplicate triples
- Remove inconsistent URIs
- Ensure consistent predicate alignment
- Remove unwanted literal-heavy predicates (if necessary)
- Ensure graph connectivity (avoid isolated components)

Why These Volume Constraints Matter

KB Size Consequence

< 10k triples Embeddings collapse to frequency effects 50k--200k triples Stable training and meaningful structure > 500k triples High memory and computation cost

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Final Deliverables

You need to prepare the following files for next lab session and final submission.

1. **Expanded KB file** (RDF/NT)
2. **Ontology file**
3. **Alignment file**
4. **Statistics report**:
    - Number of triplets
    - Number of entities
    - Number of relations
5. **Short methodological report (3--5 pages)**:
    - Alignment strategy
    - Expansion strategy
    - Challenges encountered
    - Final KB statistics

You should integrate 4 and 5 in the final report.

Next Lab Session

The resulting KB will be used for:

- Train/validation/test split
- Knowledge Graph Embedding
- Link prediction evaluation
- Model comparison (e.g., TransE, DistMult, ComplEx)

Your KB quality and structure will directly impact embedding performance.